

Large Language Models and Theory-of-Mind

An analysis with Indirect Speech Acts

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Are functional abilities emerging in LLMs?

SKILLS REQUIRED FOR SUCCESSFUL LANGUAGE USE

linguistic knowledge

Phonology, morphology, syntax

formal reasoning

logic, math

world knowledge

facts, concepts

situation modeling

discourse coherence, narrative
structure

communicative intent

pragmatics, common ground, goals

EMERGENT ABILITIES
INCLUDE SOME FORM
OF REASONING



Mahowald et al. (2023)

Enriched representations of meanings

Pragmatics transforms **literal meanings** into **intended meaning**

Understanding **intended meaning**
involves a **Theory of Mind (ToM)**

Correlation between the pragmatic component
of language and ToM

Related work

A fine-grained comparison of pragmatic language understanding in humans and language models

Jennifer Hu¹, Sammy Floyd^{1,2}, Olessia Jouravlev³, Evelina Fedorenko^{1,4}, Edward Gibson¹

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Chiara Barattieri di San Pietro, Federico Frau, Veronica Mangiaterra, Valentina Bambini

The pragmatic profile of ChatGPT: Assessing the communicative skills of a conversational agent

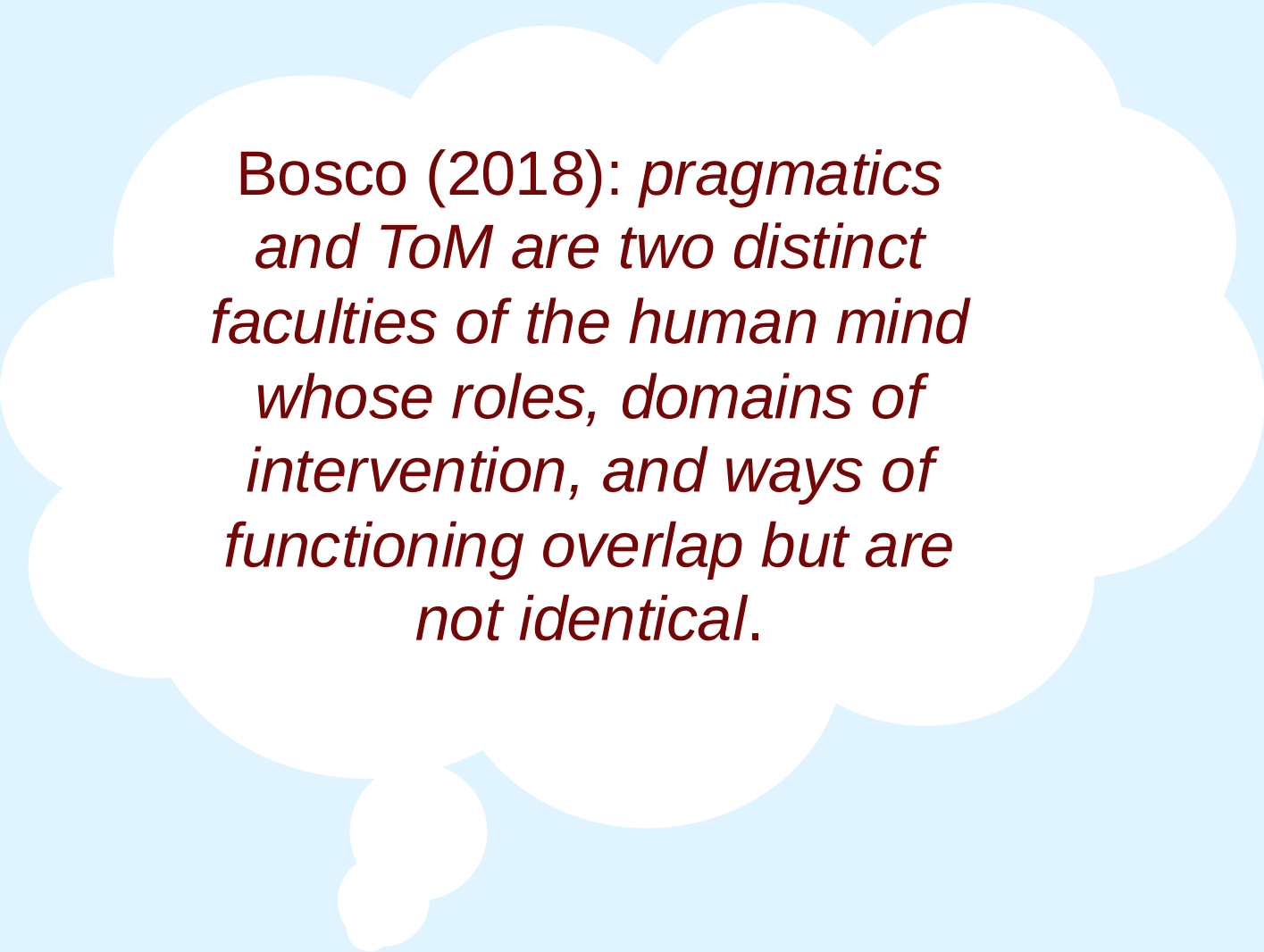
(doi: 10.1422/108136)

Sistemi intelligenti (ISSN 1120-9550)

Fascicolo 2, agosto 2023

The elaboration of pragmatic tasks does not always require mentalizing abilities

The relations between Pragmatics and Implicatures



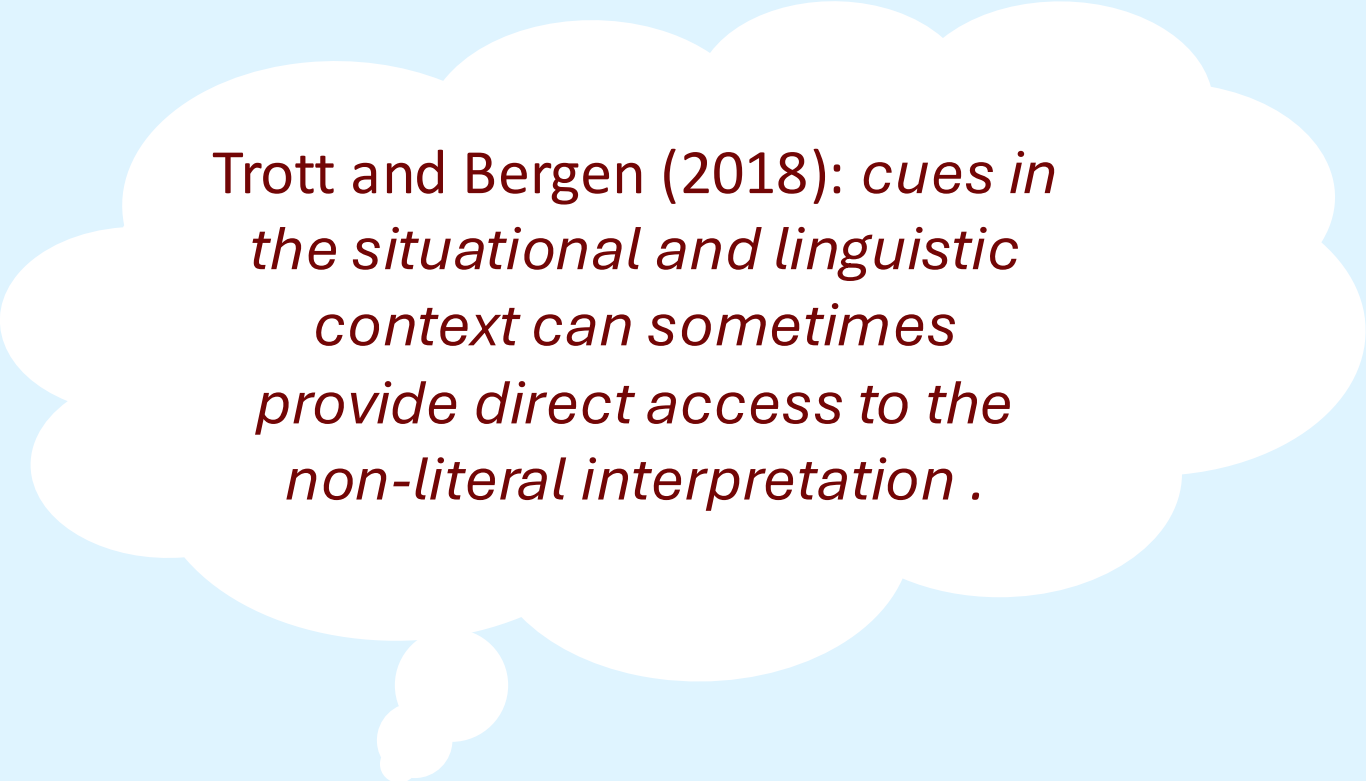
Bosco (2018): *pragmatics and ToM are two distinct faculties of the human mind whose roles, domains of intervention, and ways of functioning overlap but are not identical.*

Conversational
Implicatures can
keep distinct
Pragmatics and ToM

Why Indirect Speech Acts (ISAs)?

ISAs require mentalizing for their non-compositional nature and context-dependence

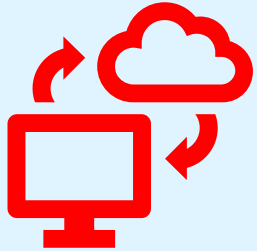
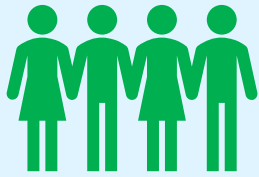
Many ISAs are
conventionalized and
idiomatic



Trott and Bergen (2018): *cues in the situational and linguistic context can sometimes provide direct access to the non-literal interpretation .*

The experiment in a nutshell

False-Belief inspired design



Obstacle to fulfilling a possible request



Interpretation depends on reasoning about beliefs and intentions (mentalizing)



pISA: benchmark to prompt ISAs

Indirect Requests (IR)

Extracted from Bergen and Trott(2018) → Old stimuli

“Can you turn on the heater?”

Indirect Requests (IR)

New stimuli

“Can you shut the front door?”

Indirect Suggestions (IS)

“Why don’t you eat your meal?”

Indirect Declinations (ID)

“Do you want some more dessert?”

“I’m full”

Indirect Threats (IT)

“You will pay for this”

Every story has two interpretations, ...

LITERAL

Mark and his friend Kyle just returned home after a month-long trip to Europe.

During their absence, they asked a friend to water the plants in their apartment and to open the windows once a week for air.

As soon as they enter, they notice that the window handle is broken, and the window is locked.

Kyle asks, "Can you open the window?".

Mark and his friend Kyle just returned home after a month-long trip to Europe. During their absence, they asked a friend to water the plants in the apartment and to open the windows once a week for air. As soon as they enter, they notice that there is a bad smell in the living room. Mark approaches the window while Kyle checks the plants.

Kyle doesn't notice that the handle is broken and asks, "Can you open the window?".

NON-LITERAL

...and two context types (long vs. short)

LITERAL

Mark and his friend Kyle just returned home.

Both immediately notice that the window handle is broken.

Kyle asks, "Can you open the window?".

Mark and his friend Kyle just returned home.

While Kyle is checking the plants, Mark immediately notices that the handle is broken, and the window is locked.

Kyle asks, "Can you open the window?".

NON-LITERAL

Task in numbers

160 Stories : 80 long context and 80 short context

40 literal and 40 non-literal (each)

Multi-choice question answering: 4 possible answers

Non Literal

Literal

Distractor

Distractor

Speakers

8 balanced
groups, each
tested on 30
speakers

240 speakers in
total

Survey questions



In this test, you will read short stories and you will have to answer a question about one character of the story. READ EACH STORY CAREFULLY! You must choose one among 4 possible answers. Choose the answer that you think to be correct given the story.

The test contains three check questions. IMPORTANT. The participant will not be paid unless it answers to these questions correctly.

Mark and his friend Kyle just returned home after a month-long trip to Europe. During their absence, they asked a friend to water the plants in their apartment and to open the windows once a week for air.

*

As soon as they enter, they notice that there is a bad smell in the living room. Both immediately notice that the window handle is broken, and the window is locked. Kyle asks, "Can you open the window?"

What does Kyle intend to say?

- ☐ 1 Please open the window.
- ☐ 2 Are you able to open the window?
- ☐ 3 I'll try to open the window.
- ☐ 4 We should fix the window.

Prompts

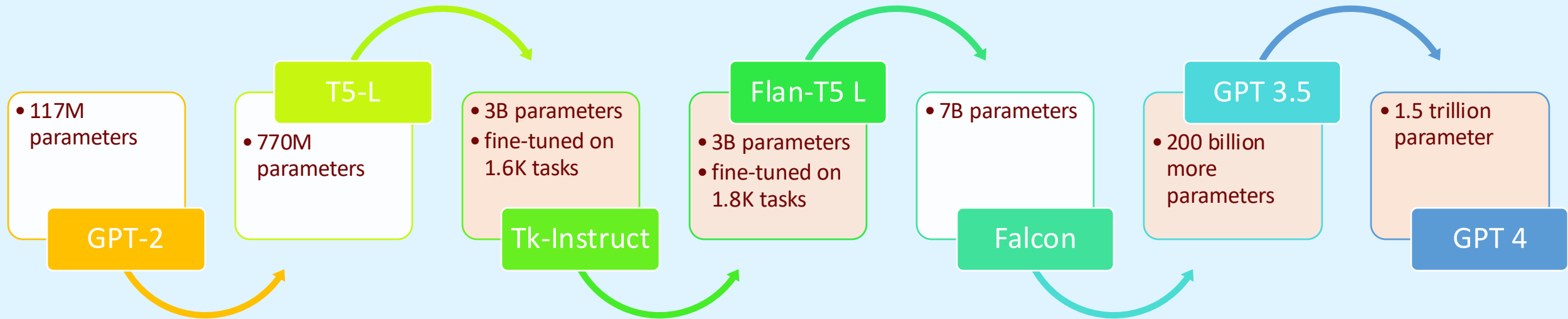
"Task: I'll give you a story and I'll ask you to answer a question about one character of the story. I'll give you four possible answers to the question and you must choose the right one. The possible answers are 1, 2, 3 and 4.

Story: I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin to pass me the glass so I can pour him more wine, but he tells me: "I need to drive".

What does Kevin intend to say?

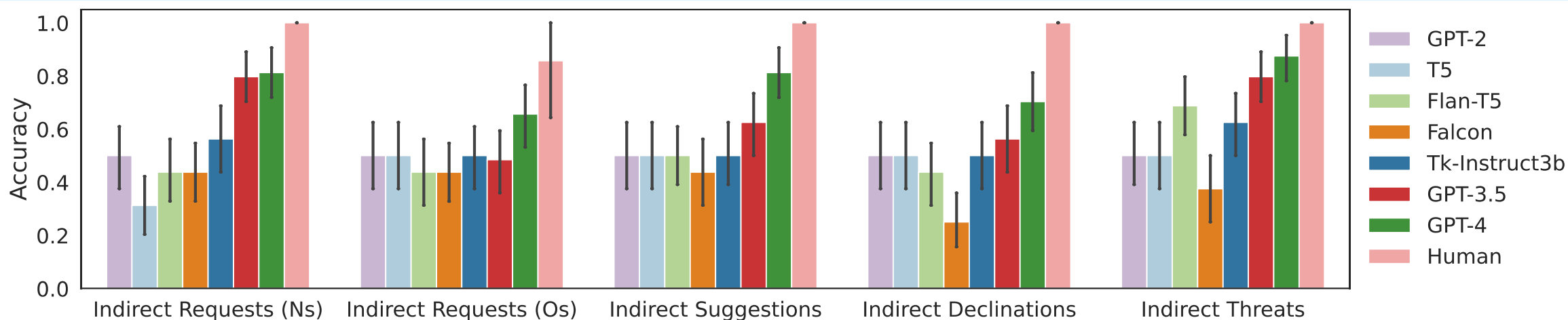
Options: 1 Please, do not pour me more wine. 2 I drive tonight 3 I don't like this wine. 4 We should drink French wine. **The correct answer is:"**

Models

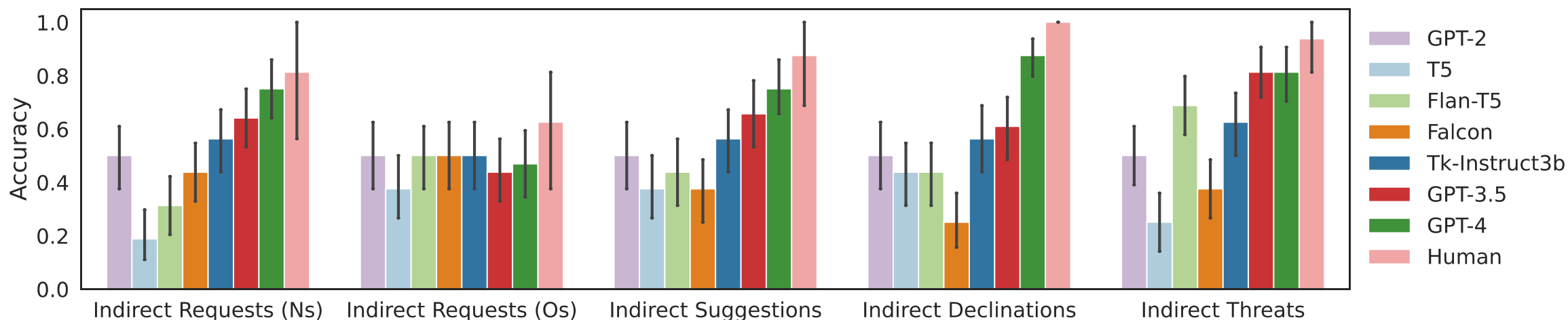


Total Accuracy

Full context

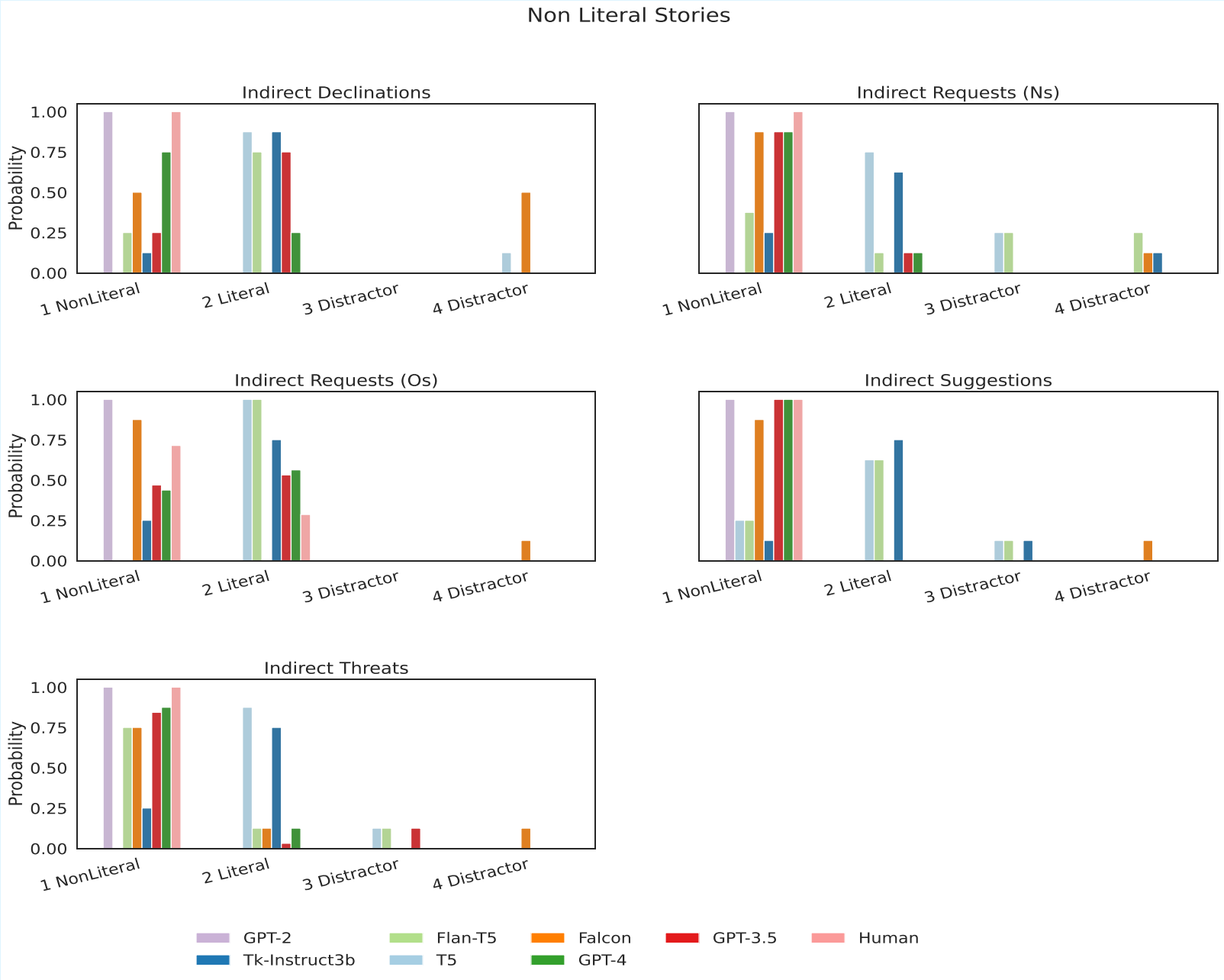


Short context

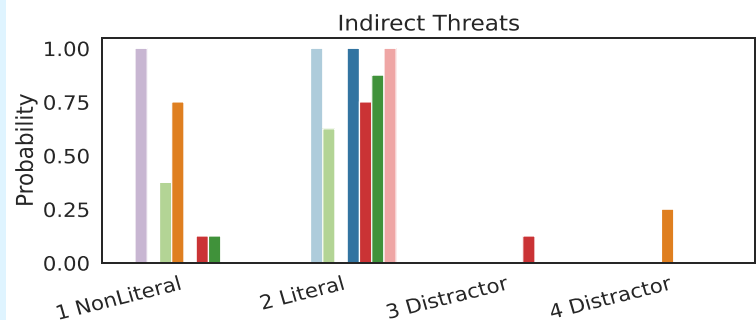
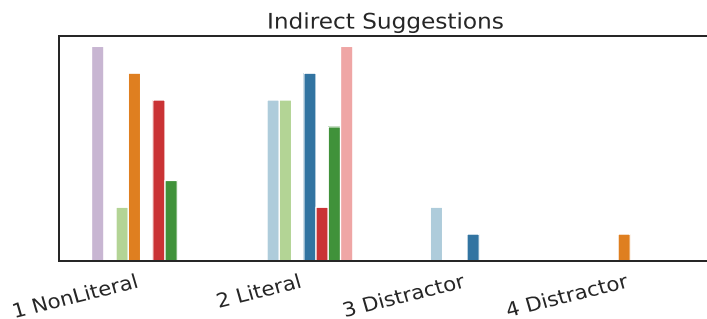
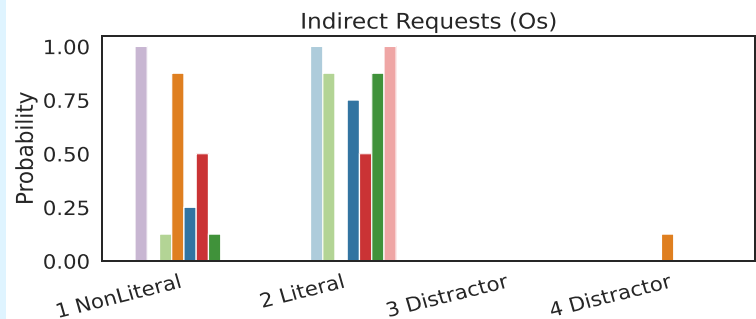
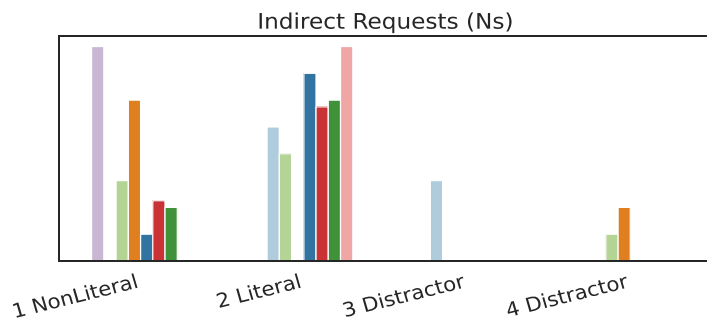
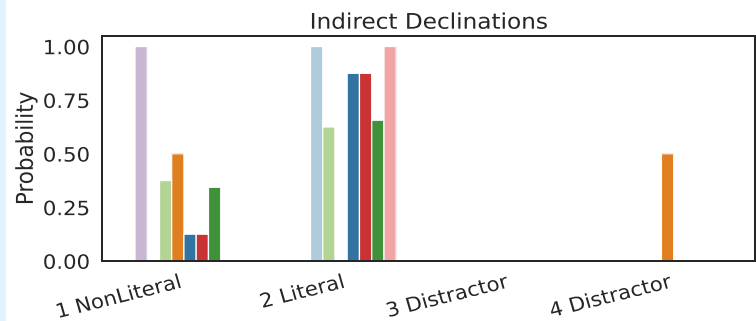


NON LITERAL STORIES

answer
distribution



Literal Stories



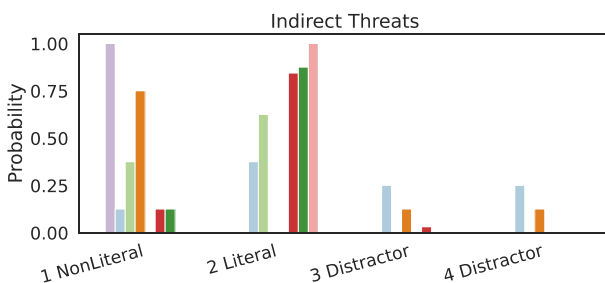
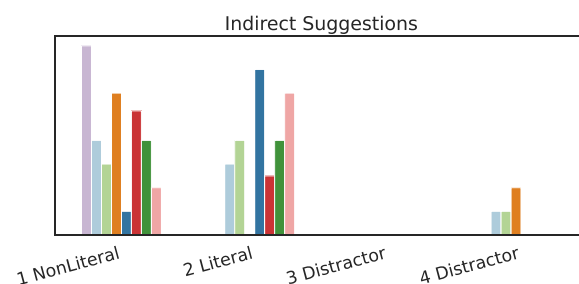
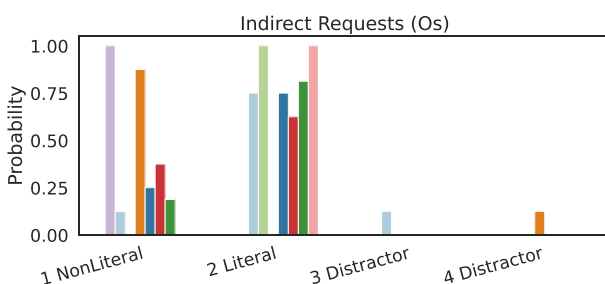
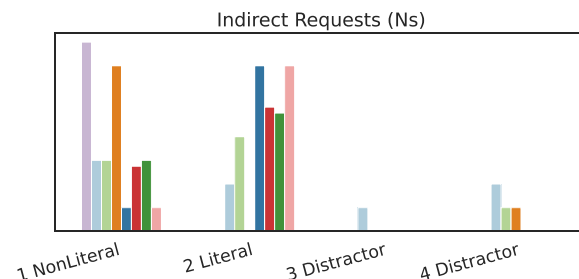
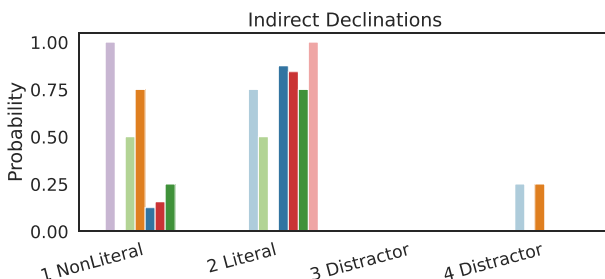
LITERAL STORIES

answer distribution

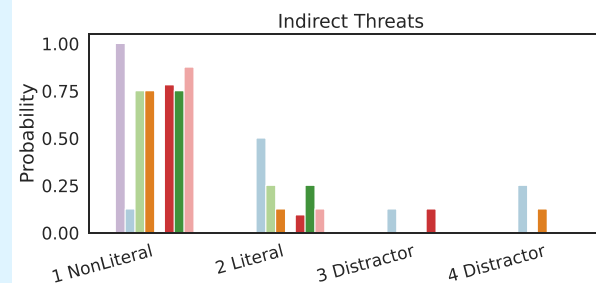
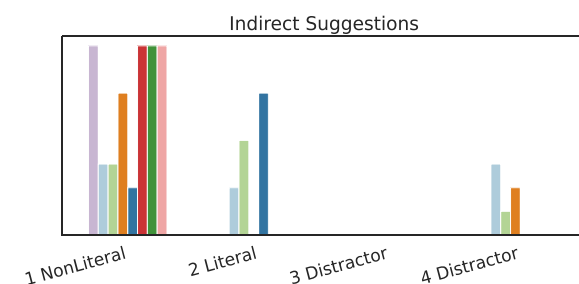
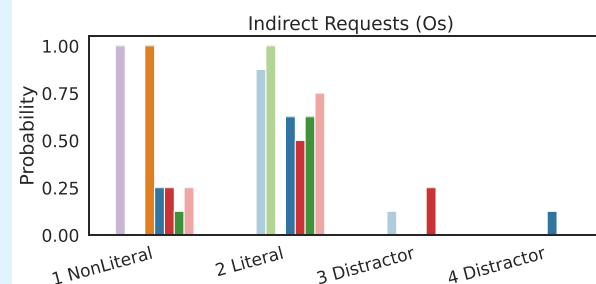
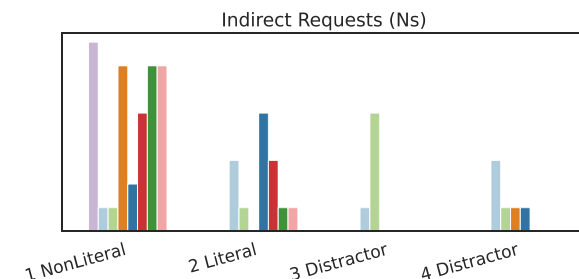
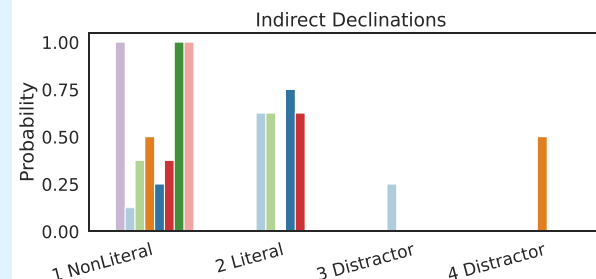
Answer distribution (short stories)

Literal Stories

Non Literal Stories

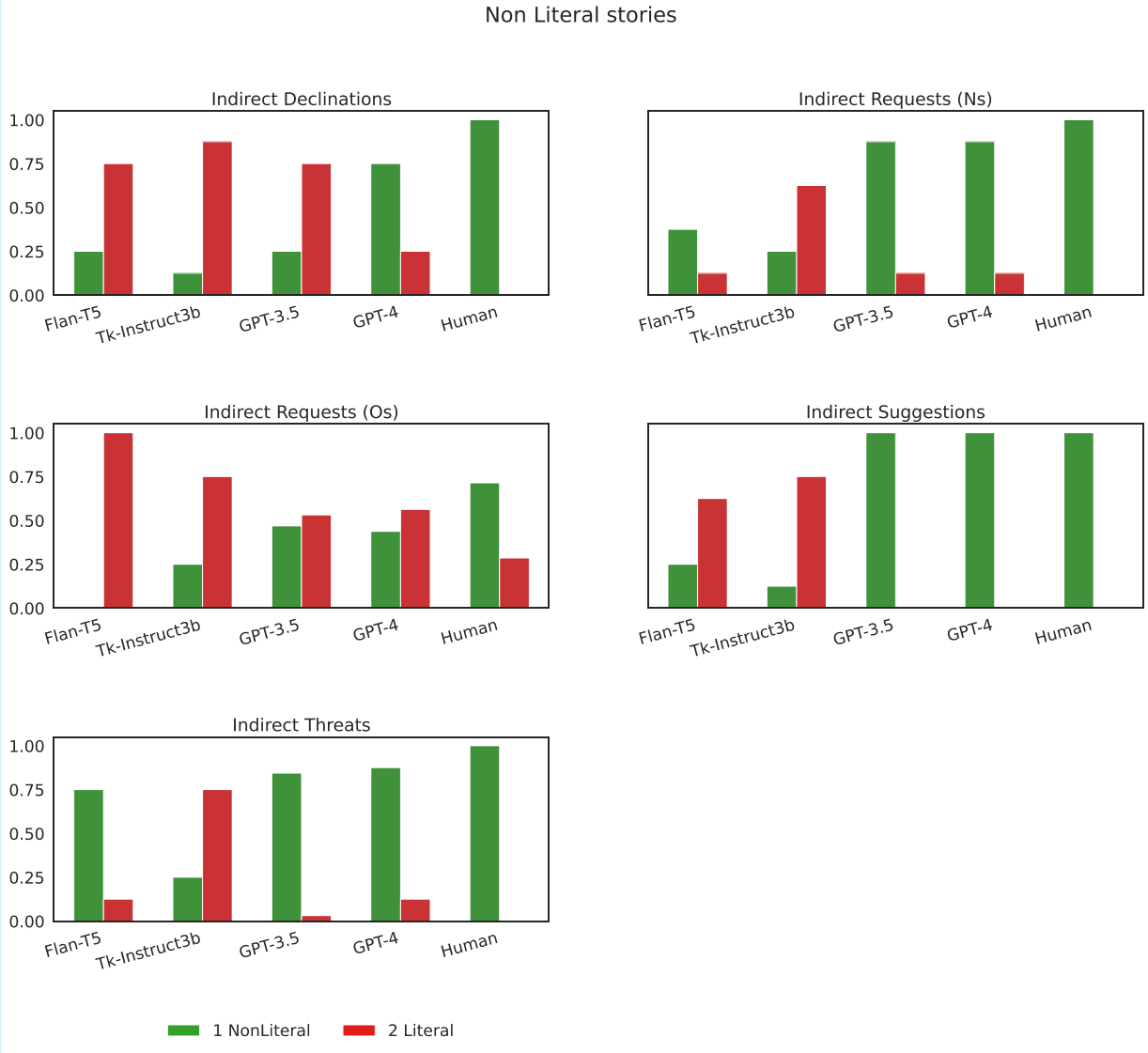
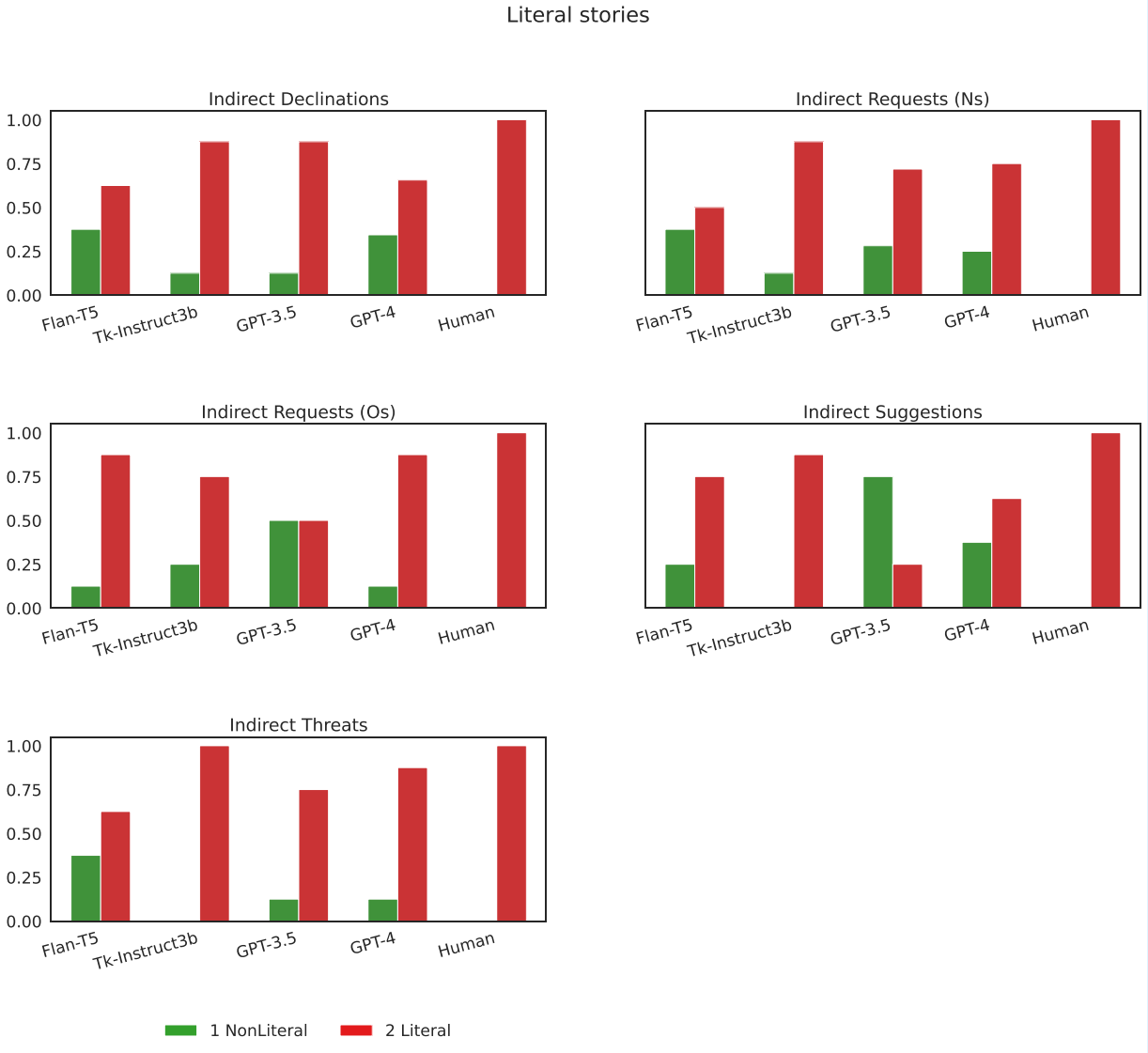


■ GPT-2
 ■ Flan-T5
 ■ Falcon
 ■ GPT-3.5
 ■ Human
 ■ Tk-Instruct3b
 ■ T5
 ■ GPT-4

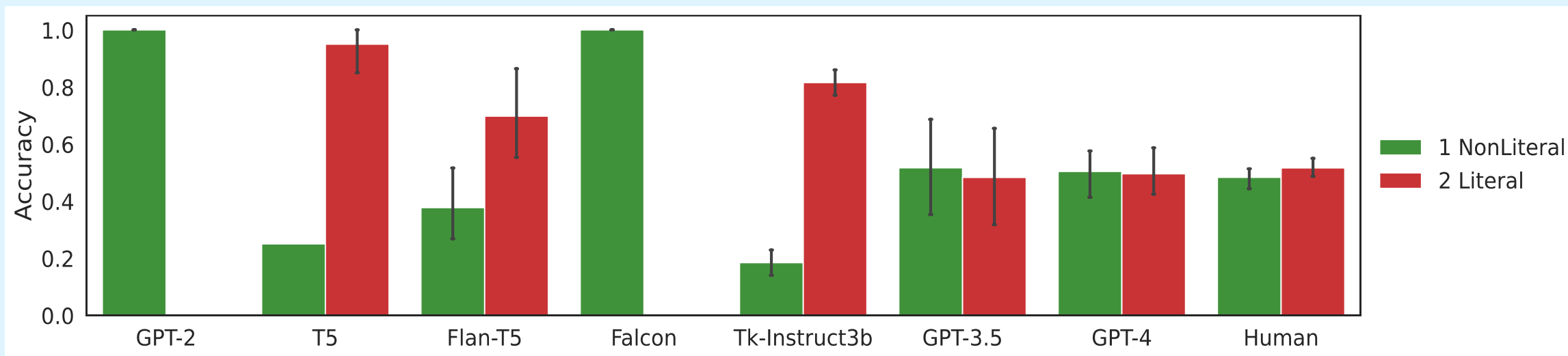


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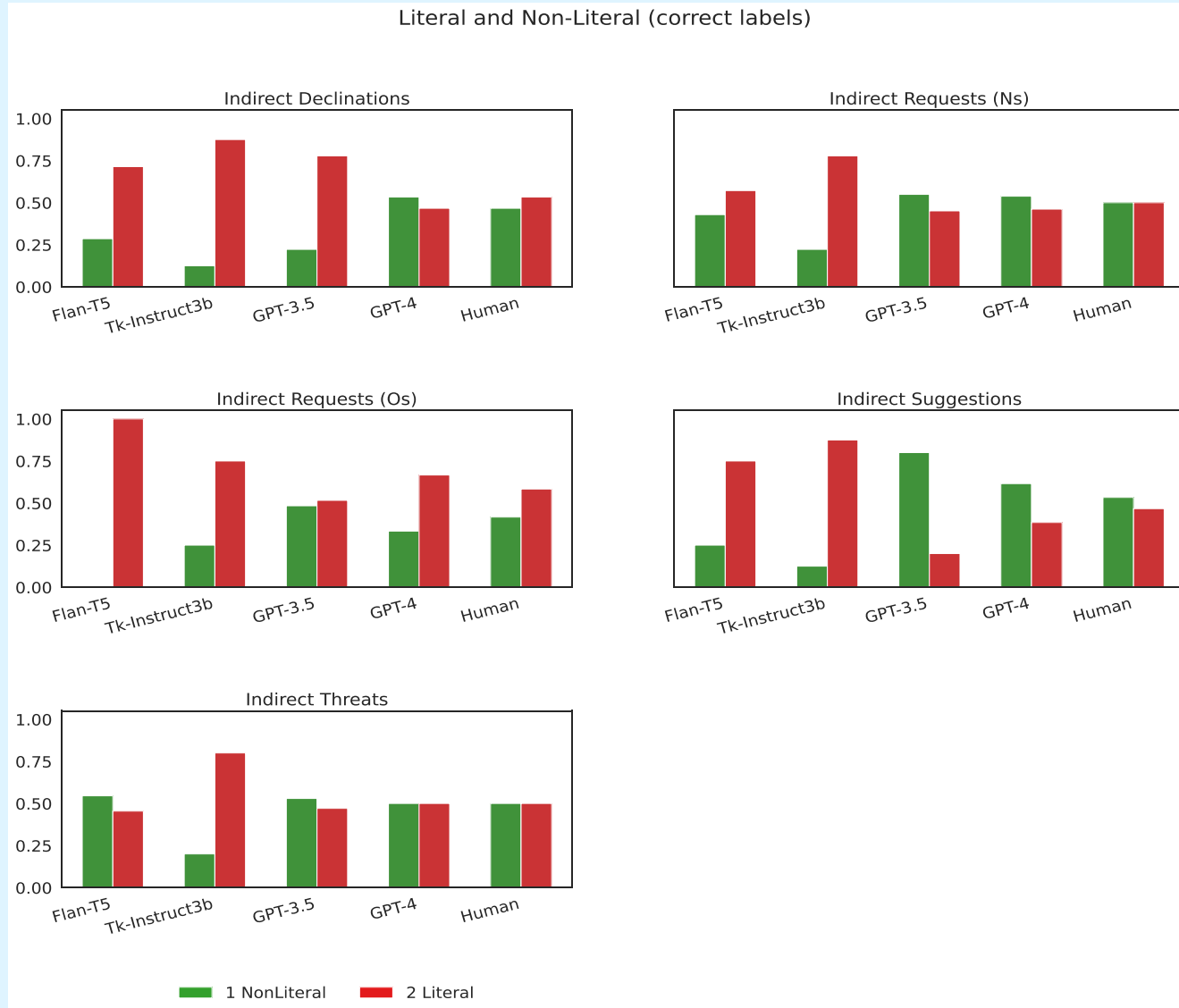
Another Detailed Comparison



Correct labels distribution



Inspecting correct labels



- Models size and instruction determine different error patterns
- Indirect Declinations appear as less conventionalized
- Indirect Requests and Indirect Suggestions select the default non-literal interpretation

Final Remarks

Does model size influence performance?

- Yes, size and instruction tuning increase accuracy

Does context influence inference?

- No, models are likely to exploit local features

Do LLMs perform a kind of ToM-based interpretation?

- No, interpretation seems to be determined from conventionalization of ISAs

Do models and humans make similar error patterns?

- Model size influences error patterns

Further Developments

Extend the experiment to Conversational Implicatures that do not require mentalizing

Include other conventionalized linguistic expression that encourage a non-literal interpretation

Develop novel methods to isolate enriched meaning representations from literals

Create a more robust benchmark to improve evaluation techniques



Thank you!